Data Base Management System Lab (BAI 201)

Department of Computer Science Engineering
Indira Gandhi Delhi Technical University for Women Kashmere Gate, Delhi

SUBMITTED BY -
Mythri Jadon (12201012024)
Naeema TN (12301012024)

4th Sem
CSE 2

SUBMITTED TO –
Ms. Tamal Misra

PROJECT REPORT

Vehicle Rental System

Database Management System | Academic Project

TABLE OF CONTENT

1. Objective of the Project

The primary objective of this project is to design and develop a **Vehicle Rental System**, a database-driven application that allows customers to easily search, browse, and book rental vehicles in a seamless and efficient manner. The system is designed to replace manual record-keeping, minimize booking errors, and provide a centralized platform for managing vehicle details, customer information, and booking records. It also aims to enhance user experience by offering features such as search, sorting, and booking tracking.

1.1 Objectives

- To design and develop a structured and efficient database system consisting of three interrelated tables: Vehicles, Customers, and Bookings, ensuring proper data management and integrity.
- To create a web-based interface that allows customers to easily search, browse, and explore available vehicles in a user-friendly manner.
- To implement filtering and sorting functionalities based on key attributes such as model, fuel type, and price per day, improving the efficiency of vehicle selection.
- To enable customers to select the rental duration and automatically calculate the total rental cost based on the selected number of days.
- To maintain accurate booking records and provide customers with the ability to view and track their current and past bookings.
- To implement a reminder mechanism for return dates in order to ensure timely vehicle returns and enhance system reliability.

1.2 Key Problems Solved

1. Eliminates manual, paper-based booking processes that are prone to errors and inefficiencies.
2. Prevents double bookings by verifying vehicle availability before confirming a reservation.
3. Avoids duplicate customer accounts through validation of unique details such as phone number and license number.
4. Provides transparent and real-time pricing based on the selected rental duration.
5. Keeps customers informed about return deadlines through an automated reminder system.

2. About the Dataset Created

The dataset for this project was custom-created to simulate a real-world vehicle rental system. It consists of three relational tables designed to capture all key aspects of the rental process, including vehicle inventory, customer details, and booking records. The tables used in this system are **Vehicles**, **Customers**, and **Bookings**, which are organized within a dataset named **vehicle_rental** for efficient data management and retrieval.

```
mysql> use vehicle_rental;
Database changed
mysql> show tables;
+--------------------------+
| Tables_in_vehicle_rental |
+--------------------------+
| bookings                 |
| customers                |
| vehicles                 |
+--------------------------+
3 rows in set (0.09 sec)
```

2.1 Vehicles Table (5,000 Records)

The Vehicles table is the core component of the system, storing detailed information about each vehicle available for rental. It includes attributes such as vehicle_id, model, type, fuel_type, seating_capacity, registration_no, price_per_day, and availability. The vehicle_id serves as the primary key, uniquely identifying each vehicle in the dataset.

The dataset has been synthetically generated to simulate real-world rental data, with entries representing a wide variety of vehicles such as bikes, cars, and SUVs. Sample records include vehicles like *Yamaha R15, Royal Enfield Classic 350, Honda City, Tata Nexon, and Toyota Innova*, ensuring diversity in vehicle types and pricing.

The availability field is stored as a Boolean value (0 or 1), where 1 indicates the vehicle is available for booking and 0 indicates it is currently unavailable.

```
mysql> desc vehicles;
+-------------------+--------------+------+-----+---------+-------+
| Field             | Type         | Null | Key | Default | Extra |
+-------------------+--------------+------+-----+---------+-------+
| vehicle_id        | int          | NO   | PRI | NULL    |       |
| model             | varchar(100) | YES  |     | NULL    |       |
| type              | varchar(50)  | YES  |     | NULL    |       |
| fuel_type         | varchar(50)  | YES  |     | NULL    |       |
| seating_capacity  | int          | YES  |     | NULL    |       |
| registration_no   | varchar(50)  | YES  |     | NULL    |       |
| price_per_day     | int          | YES  |     | NULL    |       |
| availability      | tinyint(1)   | YES  |     | NULL    |       |
+-------------------+--------------+------+-----+---------+-------+
8 rows in set (0.04 sec)
```

```
mysql> select * from vehicles;
```

vehicle_id	model	type	fuel_type	seating_capacity	registration_no	price_per_day	availability
1	Yamaha R15	Bike	Petrol	2	DL48AB1000	2614	1
2	Royal Enfield Classic 350	Bike	Petrol	2	DL87AB1001	2670	1
3	Honda City	Car	Electric	5	DL49AB1002	2542	1
4	Suzuki Baleno	Car	Petrol	5	DL55AB1003	731	1
5	Tata Nexon	SUV	Petrol	5	DL45AB1004	2160	1
6	Toyota Innova	SUV	Diesel	7	DL10AB1005	2193	1
7	Toyota Innova	SUV	Diesel	7	DL20AB1006	915	1
8	Honda City	Car	Diesel	5	DL44AB1007	2319	1
9	Suzuki Baleno	Car	Diesel	5	DL13AB1008	1802	1
10	Yamaha R15	Bike	Petrol	2	DL84AB1009	1508	1
11	Bajaj Pulsar 150	Bike	Diesel	2	DL83AB1010	1024	1
12	Royal Enfield Classic 350	Bike	Diesel	2	DL43AB1011	2760	1
13	Bajaj Pulsar 150	Bike	Petrol	2	DL54AB1012	816	1
14	Toyota Fortuner	SUV	Petrol	7	DL45AB1013	2562	1
15	Hyundai Creta	SUV	Petrol	5	DL5AB1014	2681	1
16	Tata Nexon	SUV	Diesel	5	DL53AB1015	2244	1
17	Honda Activa	Bike	Electric	2	DL28AB1016	1940	1
18	Honda City	Car	Electric	5	DL27AB1017	1685	1
19	Honda Activa	Bike	Petrol	2	DL87AB1018	1236	1
20	Royal Enfield Classic 350	Bike	Diesel	2	DL18AB1019	1570	0
21	Royal Enfield Classic 350	Bike	Petrol	2	DL52AB1020	503	1
22	Royal Enfield Classic 350	Bike	Petrol	2	DL13AB1021	1070	1
23	Royal Enfield Classic 350	Bike	Petrol	2	DL38AB1022	2490	1
24	Toyota Fortuner	SUV	Diesel	7	DL78AB1023	1505	1
25	Bajaj Pulsar 150	Bike	Diesel	2	DL36AB1024	2264	1
26	Toyota Fortuner	SUV	Petrol	7	DL25AB1025	1333	1
27	Yamaha R15	Bike	Petrol	2	DL27AB1026	758	1
28	Mahindra Scorpio	SUV	Electric	7	DL33AB1027	2290	1
29	Maruti Swift	SUV	Diesel	4	DL89AB1028	2015	1
30	Royal Enfield Classic 350	Bike	Diesel	2	DL30AB1029	739	1
31	Maruti Swift	Car	Diesel	4	DL88AB1030	2160	1

Dataset Characteristics:

- **Total Records:** 5,000 vehicles
- **Vehicle Types:** Bikes, Cars, SUVs (including models like Yamaha R15, Honda City, Toyota Fortuner, etc.)
- **Fuel Types:** Petrol, Diesel, Electric (as observed in dataset entries)
- **Seating Capacity:** Typically ranges from 2 (bikes) to 7 (SUVs like Innova/Fortuner)
- **Price per Day:** Varies dynamically (approx. ₹500 to ₹3000+ based on sample data, and higher for premium vehicles)
- **Availability:** Stored as a boolean value (0/1), updated based on booking status

2.2 Customers Table

The Customers table stores information related to users who register in the system. It includes attributes such as customer_id, name, phone, license_no, and password. The customer_id serves as the primary key, uniquely identifying each customer in the database.

This table is designed to ensure secure and organized management of customer data. Each user is assigned a unique customer ID at the time of registration. To prevent duplicate accounts, important fields such as phone number and license number are validated during the signup process.

The password field is used for authentication during login, enabling secure access to the system.

```
mysql> desc customers;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| customer_id | varchar(20)  | NO   | PRI | NULL    |       |
| name        | varchar(100) | YES  |     | NULL    |       |
| phone       | varchar(15)  | YES  |     | NULL    |       |
| license_no  | varchar(50)  | YES  |     | NULL    |       |
| password    | varchar(50)  | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
mysql> select * from customers;
+-------------+--------+------------+------------+----------+
| customer_id | name   | phone      | license_no | password |
+-------------+--------+------------+------------+----------+
| cust_1      | Naeema | 1234567890 | DL4647     | 123456   |
+-------------+--------+------------+------------+----------+
1 row in set (0.00 sec)
```

Validation Rules:

- Phone number and license_no must be unique to prevent duplicate customer registrations.
- Customer ID is system-generated in a unique format (e.g., *cust_1, cust_2*) to ensure each user is uniquely identified.
- Passwords are stored securely and are not maintained in plain text, ensuring safe authentication.

2.3 Bookings Table

The Bookings table records all rental transactions between customers and vehicles. It includes details such as booking_id (primary key), customer ID, vehicle ID, rental date, return date, total price, and booking status.

This table links customers to vehicles for a specific rental period, allowing the system to track active and completed bookings and support billing operations.

The status field indicates the booking state (e.g., Confirmed or Cancelled), with a default value of *Confirmed*. The total_price represents the overall rental cost, which can be calculated based on rental duration and vehicle pricing.

```
mysql> desc bookings;
+-------------+---------------+------+-----+-----------+-------+
| Field       | Type          | Null | Key | Default   | Extra |
+-------------+---------------+------+-----+-----------+-------+
| booking_id  | varchar(50)   | NO   | PRI | NULL      |       |
| customer_id | varchar(50)   | YES  |     | NULL      |       |
| vehicle_id  | varchar(50)   | YES  |     | NULL      |       |
| rental_date | date          | YES  |     | NULL      |       |
| return_date | date          | YES  |     | NULL      |       |
| total_price | decimal(10,2) | YES  |     | NULL      |       |
| status      | varchar(20)   | YES  |     | Confirmed |       |
+-------------+---------------+------+-----+-----------+-------+
7 rows in set (0.00 sec)
```

```
mysql> select * from bookings;
+------------+-------------+------------+-------------+-------------+-------------+-----------+
| booking_id | customer_id | vehicle_id | rental_date | return_date | total_price | status    |
+------------+-------------+------------+-------------+-------------+-------------+-----------+
| BID_1      | cust_1      | 1          | 2026-04-19  | 2026-04-20  |     2614.00 | Cancelled |
| BID_2      | cust_1      | 20         | 2026-04-23  | 2026-04-25  |     3140.00 | Confirmed |
+------------+-------------+------------+-------------+-------------+-------------+-----------+
2 rows in set (0.00 sec)
```

Derived Fields (computed at application level):

Although total_price is stored, the following values can be computed:

- **Number of Rental Days** = return_date − rental_date
- **Total Cost** = Number of Rental Days × price per day
- **Return Reminder**: Triggered when current_date ≥ return_date − 1 day

3. Entity-Relationship (ER) Diagram

3.1 ER Diagram Description

The Entity–Relationship (ER) diagram for the Vehicle Rental System consists of three main entities: Vehicles, Customers, and Bookings**,** along with their relationships.

Entity	Attributes	Relationship
VEHICLES	vehicle_id (PK), model, type, fuel_type, seating_capacity, registration_no (UQ), price_per_day, availability	One vehicle can be associated with multiple bookings (1:N with BOOKINGS)
CUSTOMERS	customer_id (PK), name, phone (UQ), license_no (UQ), password	One customer can make multiple bookings (1:N with BOOKINGS)
BOOKINGS	booking_id (PK), customer_id, vehicle_id, rental_date, return_date, total_price, status	Acts as a linking entity between CUSTOMERS and VEHICLES

3.2 Relationship Summary

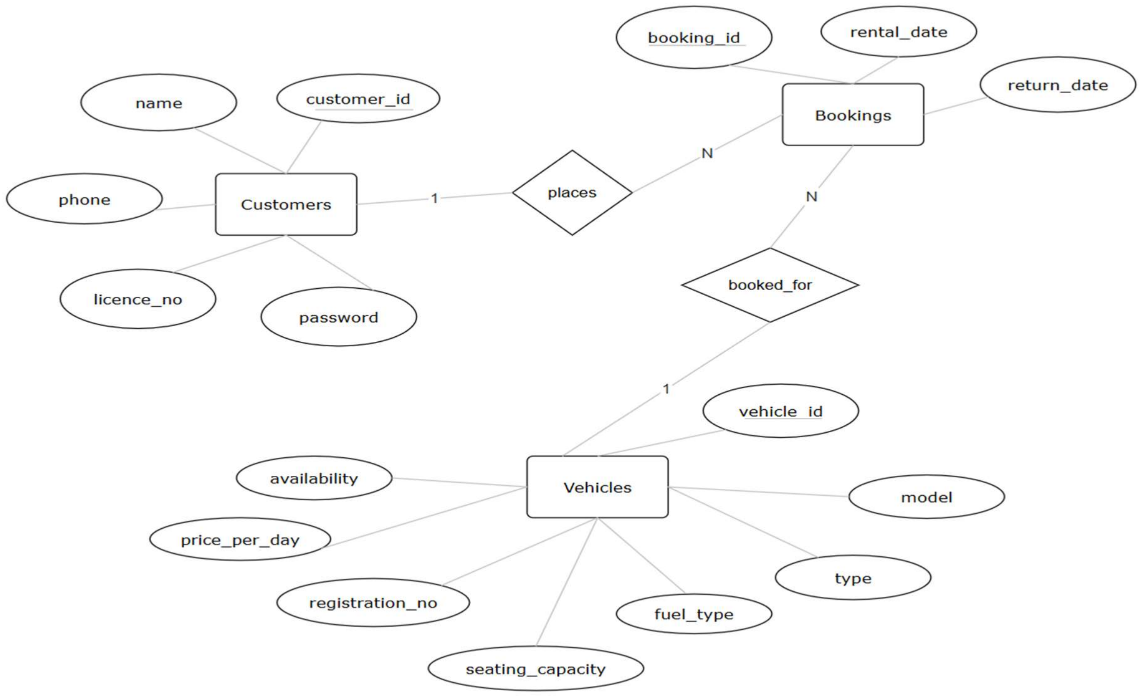
- CUSTOMERS (1) → (N) BOOKINGS (A customer can place multiple bookings)
- VEHICLES (1) → (N) BOOKINGS (A vehicle can be booked multiple times)
- BOOKINGS acts as a bridge (associative) table resolving the many-to-many relationship between CUSTOMERS and VEHICLES
- customer_id and vehicle_id in BOOKINGS logically reference their respective tables

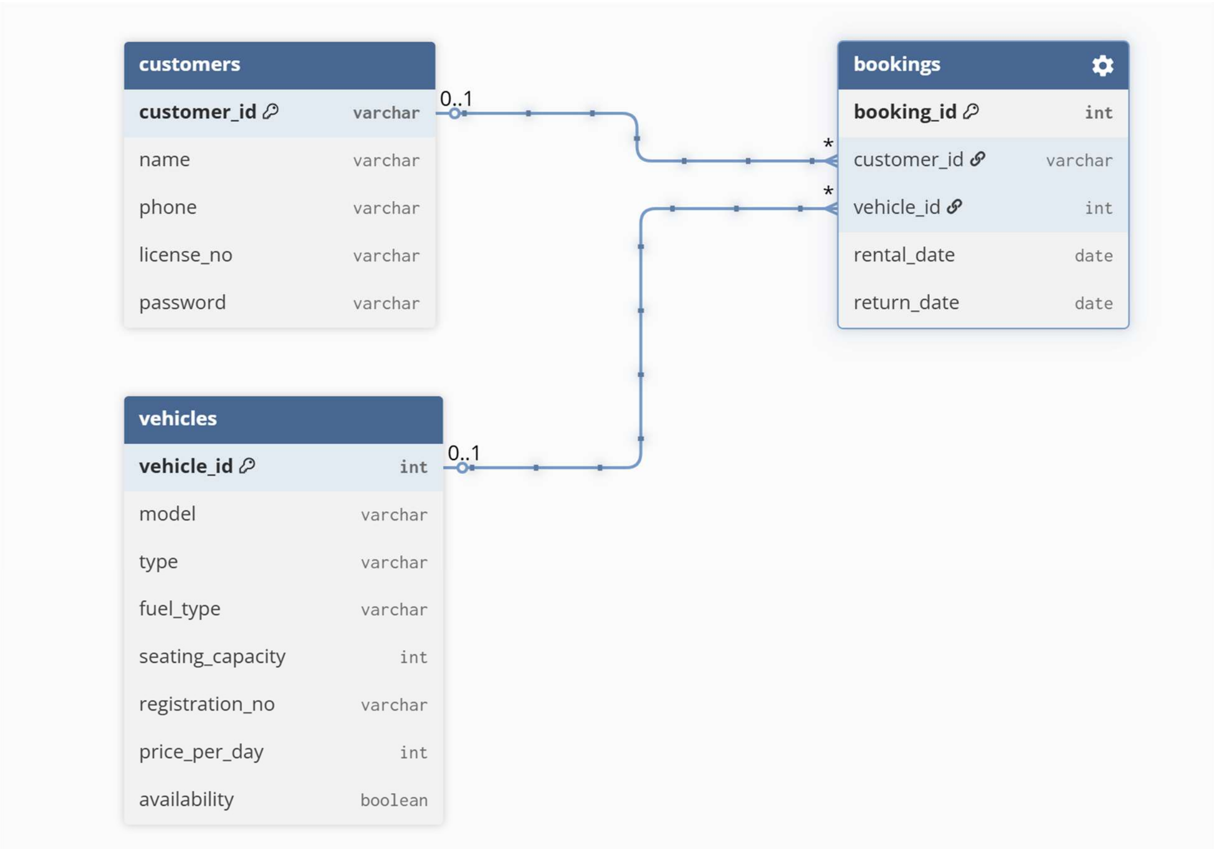
3.3 Cardinality

- One Customer → Many Bookings
- One Vehicle → Many Bookings
- Each Booking → Exactly one Customer and one Vehicle

3.4 ER Diagram

Below is a structured textual ER representation. (Replace this section with the actual ER diagram image from your project tool such as MySQL Workbench, draw.io, or dbdiagram.io)


booking_id
rental_date
return_date
Bookings
name
customer_id
phone
Customers
1
places
N
N
booked_for
licence_no
password
1
vehicle_id
availability
Vehicles
model
price_per_day
type
registration_no
fuel_type
seating_capacity


customers
customer_id varchar
name varchar
phone varchar
license_no varchar
password varchar
bookings
booking_id int
customer_id varchar
vehicle_id int
rental_date date
return_date date
0..1
*
*
vehicles
vehicle_id int
model varchar
type varchar
fuel_type varchar
seating_capacity int
registration_no varchar
price_per_day int
availability boolean
0..1

4. Workflow of the Project

The Vehicle Rental System follows a structured workflow from user registration to booking completion and return. The workflow is described in sequential phases below.

Step 1: Register (Sign-Up)

1. Open the website and navigate to the Sign-Up page.
2. Enter Name, Phone Number, Driving Licence Number, and Password.
3. If the phone number or licence number already exists, an error message is displayed.
4. If the details are valid, registration is successful.
5. A unique Customer ID is generated and shown to the user.

Step 2: Login

6. Go to the Login page.
7. Enter Customer ID and Password.
8. If the Customer ID is invalid, an error is shown.
9. If the password is incorrect, an error is displayed.
10. If credentials are correct, the user is successfully logged in.

Step 3: Search for Vehicles

11. After login, the Home Page is displayed.
12. Use the search bar to enter keywords (vehicle model).
13. The system displays available vehicles matching the search.
14. Use sort and filter options (vehicle type, fuel type, and sort by price) to refine results.

Step 4: Select and Book a Vehicle

15. Click on a vehicle to view its details.
16. Select rental duration (start date and number of days).
17. The total cost is automatically calculated.
18. Click on **Confirm Booking**.
19. The booking is saved, and the vehicle becomes unavailable.
20. A confirmation page is shown with Booking ID and details.

Step 5: Track Bookings

21. Navigate to the My Bookings section.

22. View all active and past bookings.
23. Check details such as vehicle, dates, cost, and status.

Step 6: Profile & Other Pages

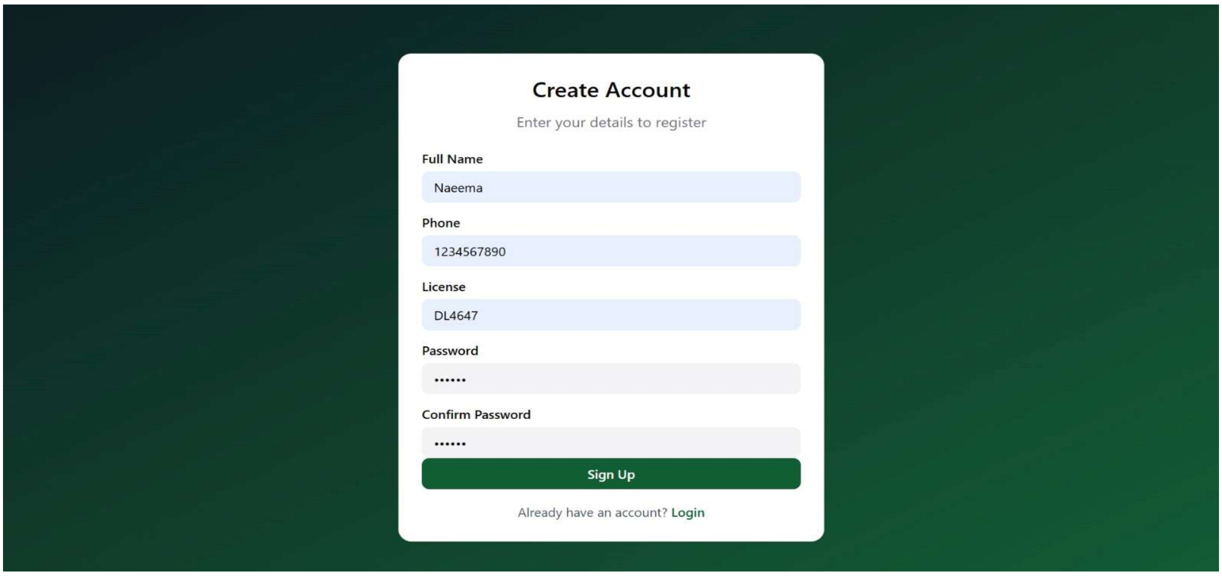
26. Visit the Profile page to view personal details and total spending.
27. Use the About Us page to learn more about the system.

5. Screenshots of the Executed Project

Below are the descriptions and placeholders for each major screen of the Vehicle Rental System. Please replace each placeholder box with the actual screenshot from your executed application.

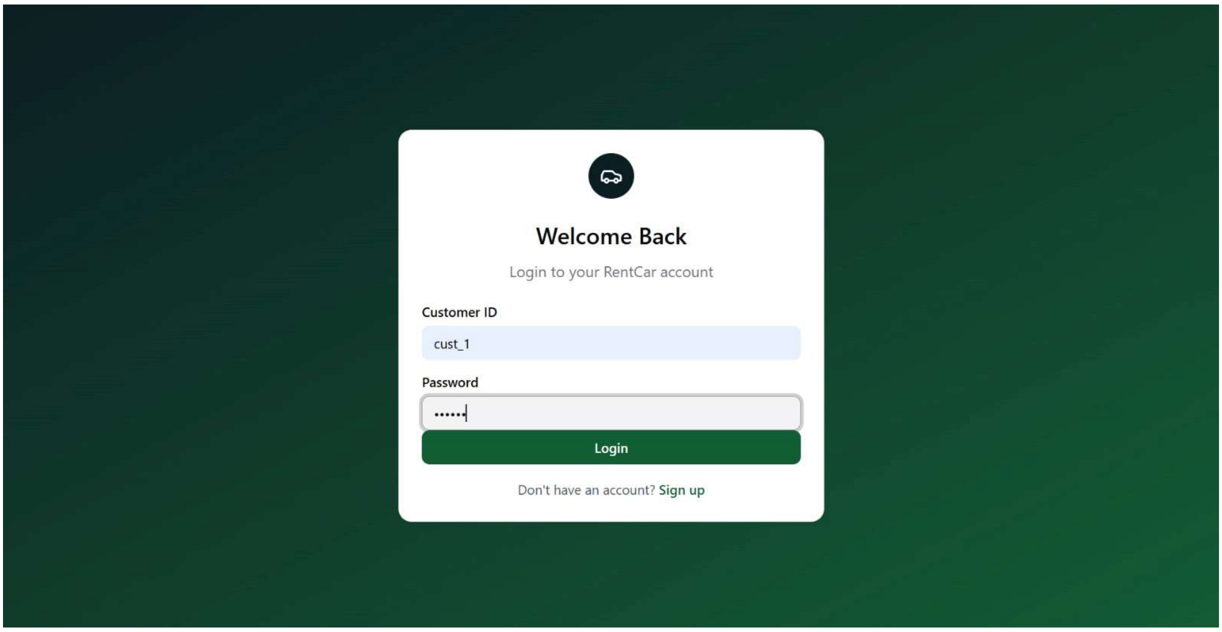
5.1 Sign-Up Page

Shows the registration form with fields for Full Name, Phone Number, Driving Licence Number, and Password. Displays an error if the user already exists.



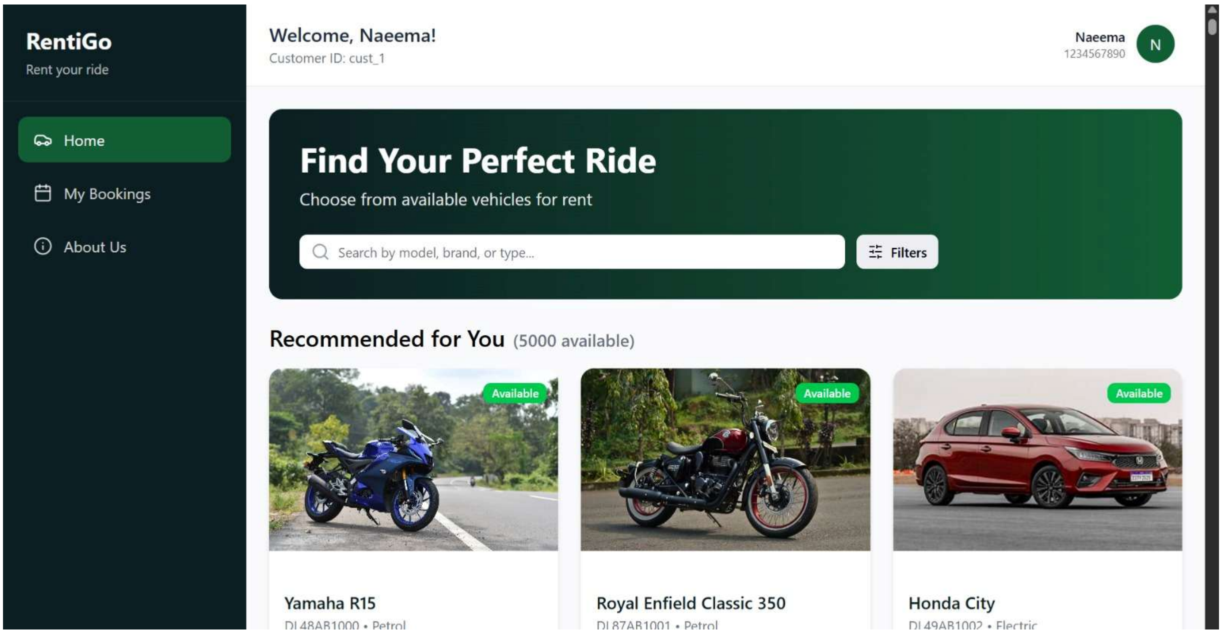
5.2 Login Page

Shows Customer ID and Password fields. Displays error messages for invalid credentials.



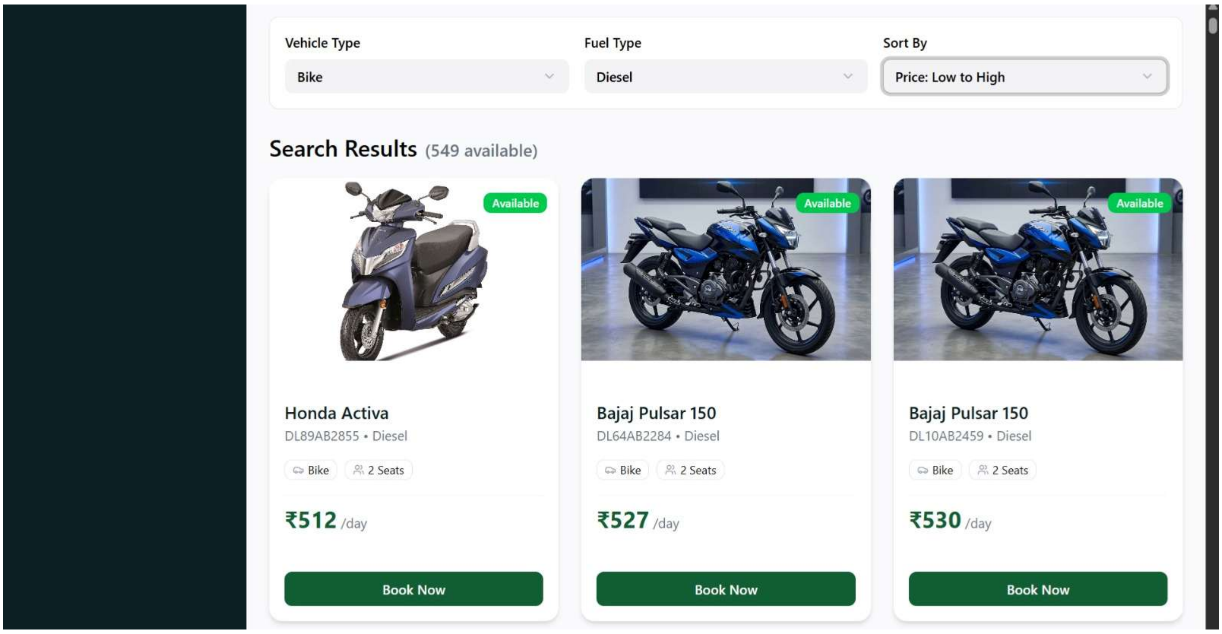
5.3 Home Page with Search Bar

Shows the search bar at the top, recommended vehicles below, and navigation links.



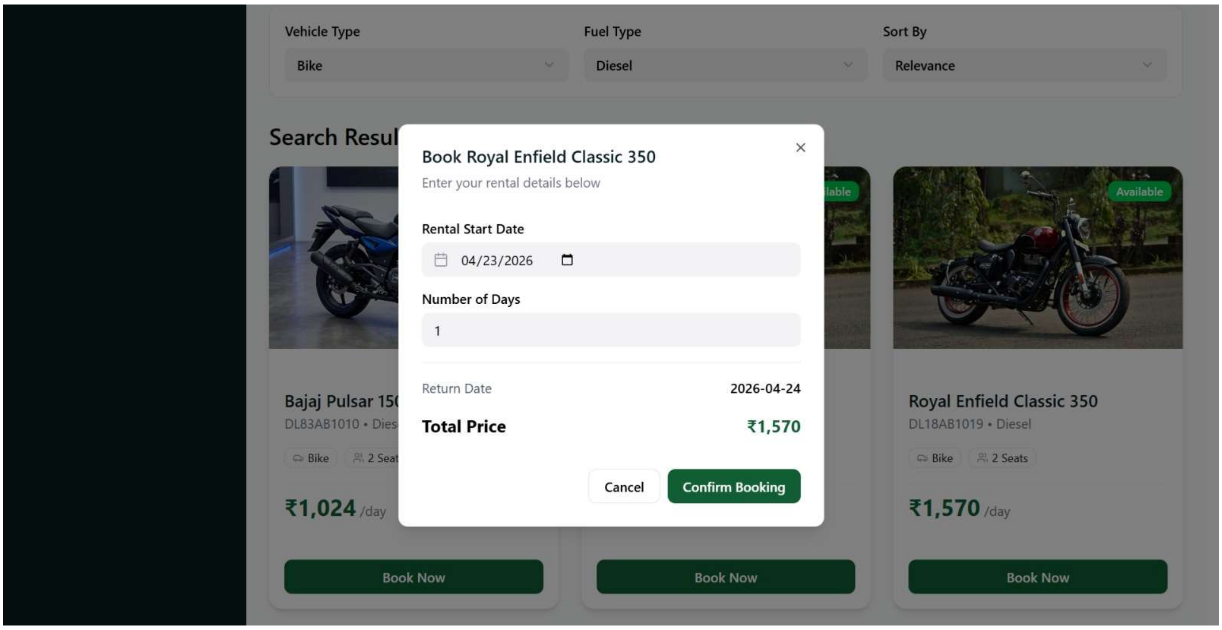
5.4 Search Results with Sort Options

Shows filtered vehicle cards after a search query, with sort dropdown (Vehicle Type, Fuel Type, Sort by).



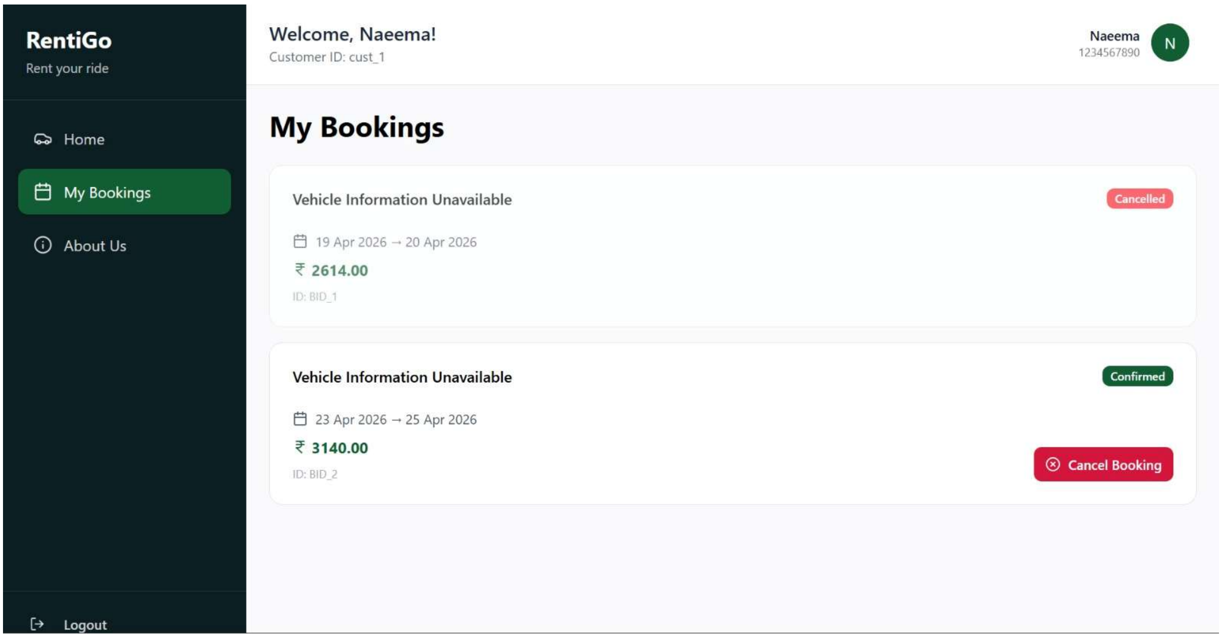
5.5 Vehicle Detail & Booking Form

Shows full vehicle details, rental start date, number of days, dynamically updated total cost, return date displayed and a Confirm Booking button.



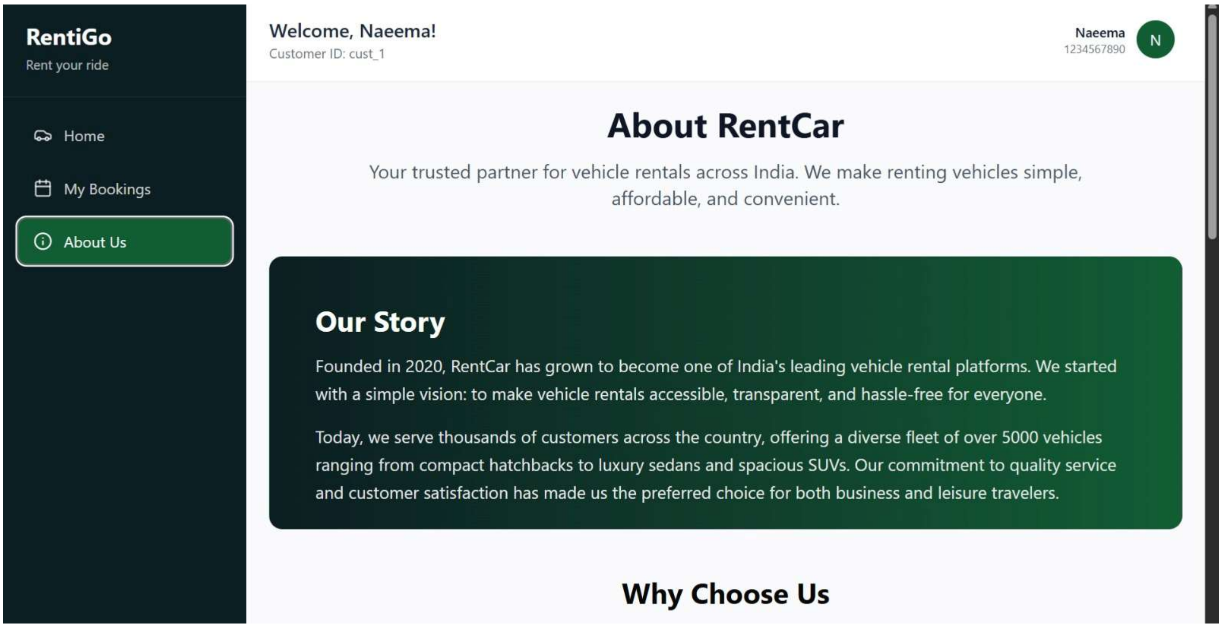
5.6 Booking Confirmation and Tracking Page

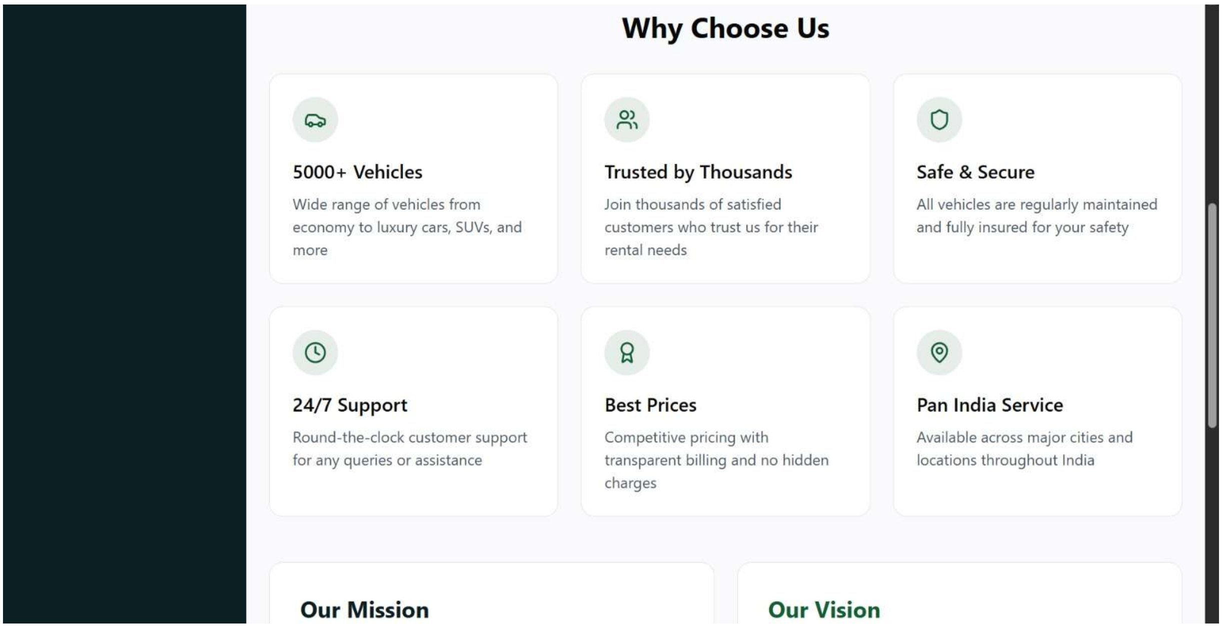
Shows the Booking ID, vehicle details, rental and return dates, and total payable amount. Shows a list of all past and active bookings for the logged-in customer, with statuses.



5.8 About Us

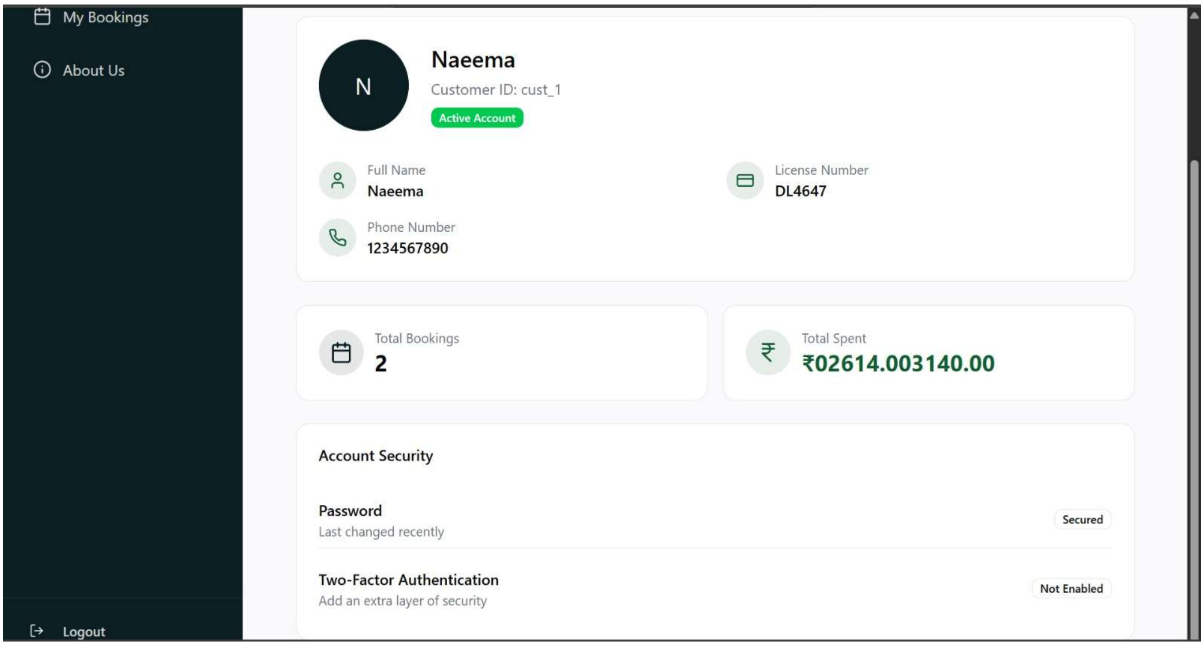
Shows about the site, what is it used for, about the site.





5.9 Profile

This consists details of the customer, name, phone number, licence number, total spent, and password.



6. Conclusion

The Vehicle Rental System successfully demonstrates the integration of a structured relational database with an application-level workflow to provide an efficient and user-friendly rental solution. The three-table schema — Vehicles, Customers, and Bookings — effectively represents the system entities and their relationships while maintaining data integrity through appropriate constraints.

Key features of the system include:

- Secure user authentication with duplicate prevention and password hashing
- Efficient search and sorting of vehicle records in real time
- Transparent booking process with automatic cost calculation
- Booking tracking with return date reminders for improved user experience

Overall, the project highlights the practical application of DBMS concepts such as ER modelling, normalization, SQL operations, and database integration. Future enhancements may include online payment integration, GPS-based tracking, and an admin dashboard for better system management.